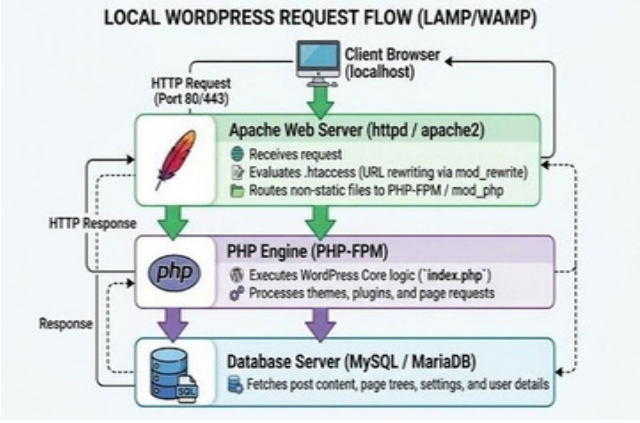
Sindhudurg Shikshan Prasarak Mandal's College of Engineering, Kankavli

A/P-HarkulBudruk,Tal- Kankavli. Dist-Sindhudurg.Maharashtra.416602
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Departments of
Computer Engineering, Computer Science&Engineering,Computer Science & Engineering (AIML)

Experiment No - 9		
Year: 2026-27	Class/Div.-TE	SEM: V
Course Code: 2115311	Course Name: Open Source Technology	
Branch: Computer Engineering, Computer Science & Engineering & AIML (Computer Science Engineering)		

Aim	Web Deployment Install and verify a running Apache web server locally; Setup a functional multi-page site by installing WordPress on top of a local LAMP/WAMP server stack.
Software Requirements	**CPU:** Dual-core 2.0 GHz or higher. **RAM:** 4 GB minimum (8 GB recommended for smoother local server performance). **Disk Space:** 2–5 GB available space (for OS, local server stack binaries, database, and media uploads).
Theory	Understanding how the components interoperate within a local development stack clarifies how local requests are handled:  **Client Request:** The web browser sends an HTTP request to http://localhost onPort 80. **Apache HTTP Server:** Acts as the front-facing web server. It listens on port 80, serves static assets (CSS, JS, images) directly, and passes dynamic requests (.php files) to the PHP processor. It uses mod_rewrite to convert clean post titles into queryable routes (e.g., /about-us → index.php?pagename=about-us).

PHP Interpreter: Parses WordPress code, runs hooks/actions, and queries the database for site data.
MySQL/MariaDB Database: Stores all site metadata, dynamic page text, configurations, user credentials, and plugin configurations across relational tables.

Procedure

Installation & Setup Procedure

Choose Option A for Windows (using XAMPP) or Option B for Linux (using native LAMP packages).

Option A: WAMP Setup on Windows (via XAMPP)
Step1:Installand Verify Apache

1. Download **XAMPP**(withP HP8.x) and execute the installer.
2. Select **Apache** and **MySQL**modules during installation.
3. Launch the**XAMPP Control Panel** and click **Start** for both Apache and MySQL.
4. **Verification:** Open your browser and navigate to http://localhost. If XAMPP's dashboard loads, Apache is correctly installed and listening on port 80.

Step 2: Create MySQL Database

1. Open http://localhost/phpmyadmin in your browser.
2. Click **Databases** → Name it wordpress_db.
3. Select utf8mb4_unicode_ci as collation and click **Create.**

Step 3: Install WordPress

1. Download the latest WordPress package from wordpress.org.
2. Extract the wordpress folder into your web root directory: C:\xampp\htdocs\mywebsite.
3. Navigate to http://localhost/mywebsite in your browser.
4. Enterd atabase details:
   - **Database Name:**wordpress_db
   - **Username:**root
   - **Password:** *(Leaveb lankbydefault)*
   - **Database Host:** localhost
5. Set site title, admin username,andpassword, then click **Install WordPress.**

Option B: LAMP Setup on Linux (Ubuntu/Debian)

1.Install Apache, MySQL, and PHP Stack: Prerequisite: Terminal access with sudo privileges.
Update your package repository and install the LAMP components along with required PHP modules:
Bash
sudo apt update

sudo apt install apache2 mysql-server php libapache2-mod-php php-mysql php-cli php-curl php-gd php-mbstring php-xml php-zip -y

2.Verify Apache Server Status: Ensures Apache runs automatically on boot and is listening on Port 80.
Verify Apache is running using systemctl:
Bash
sudo systemctl status apache2
Ensure mod_rewrite is enabled for WordPress clean URLs (Permalinks):
Bash
sudo a2enmod rewrite
sudo systemctl restart apache2
Verification: Open http://localhost in your browser to verify the Apache2 Default Page.

3.Create Dedicated MySQL Database & User: Creates isolated credentials for WordPress security.
Log in to MySQL as root and create the database and user:
SQL
sudo mysql -u root

CREATE DATABASE wordpress_db DEFAULT CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
CREATE USER 'wp_user'@'localhost' IDENTIFIED BY 'StrongPassword123!';
GRANT ALL PRIVILEGES ON wordpress_db.* TO 'wp_user'@'localhost';
FLUSH PRIVILEGES;
EXIT;

4.Deploy WordPress Files & Set Permissions: Place files in Apache's web root (/var/www/html/).
Download, extract, and assign proper ownership permissions to the web root:
Bash
cd /tmp
wget https://wordpress.org/latest.tar.gz
tar -xvzf latest.tar.gz
sudo mv wordpress /var/www/html/mywebsite

Assign Apache user (www-data) ownership
sudo chown -R www-data:www-data /var/www/html/mywebsite
sudo chmod -R 755 /var/www/html/mywebsite

5.Run Web Installer & Build Multi-Page Architecture: Complete installation via web interface.

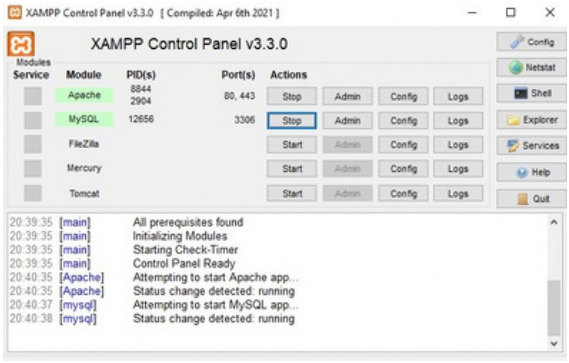
1. Visit http://localhost/mywebsite in your browser.
2. Complete the setup wizard using wordpress_db, wp_user, and StrongPassword123!.
3. Log in to the dashboard (/wp-admin).
4. **Enable Pretty Permalinks:** Go to **Settings > Permalinks**, select **Post name**, and save.

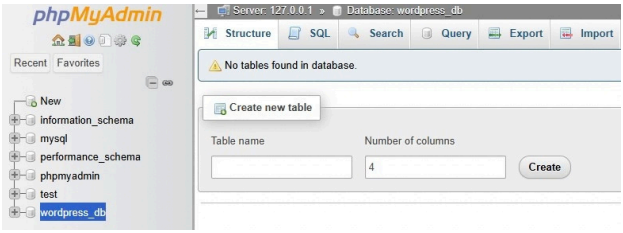
	5. **Create Pages:** Go to **Pages > Add New** and create **Home, About, Services, and Contact.** 6. **Build Menu:** Go to **Appearance > Menus**, create a "Primary Menu", add all created pages, and assign it to the Header location.
Conclusion	Local LAMP/WAMP stacks create isolated development environments by combining Apache (HTTP requests), PHP (processing logic), and MySQL (data storage). Installing WordPress on this setup provides a full multi-page site with dynamic routing and database storage, allowing you to safely test theme, plugin, and structural edits offline without costs or risks to live traffic.

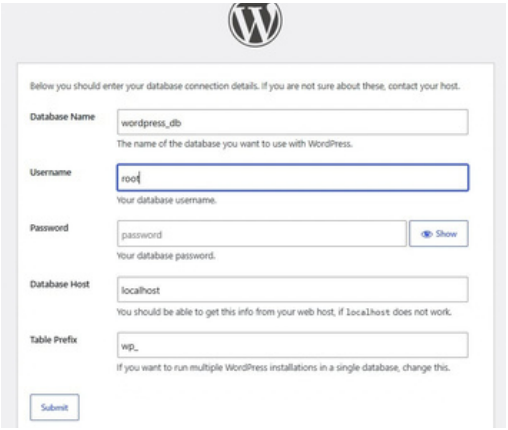
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OUTPUT







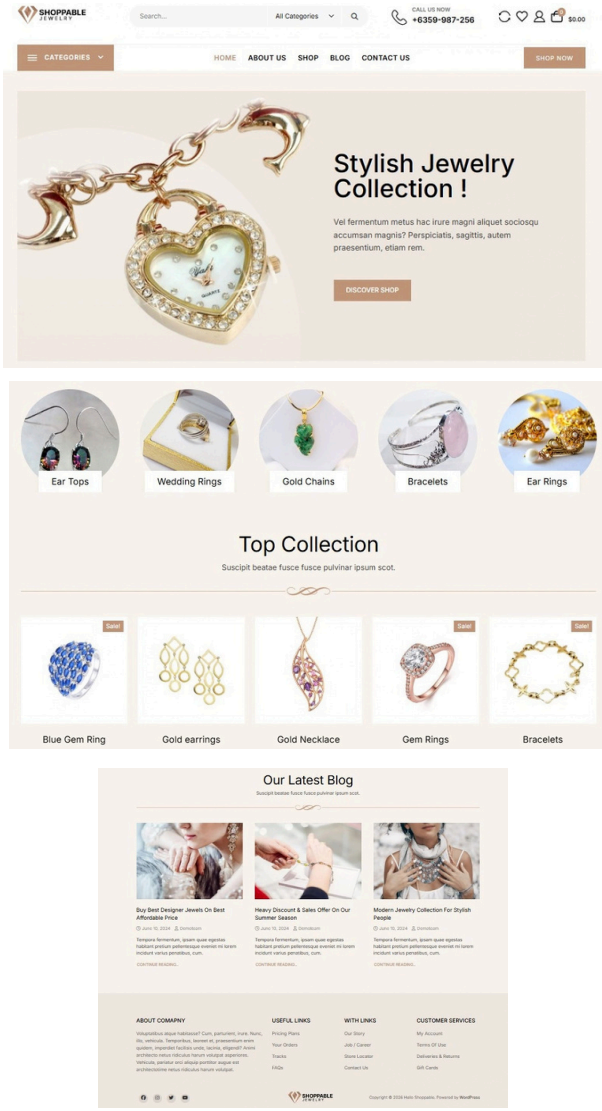
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